Name: _____ Date: _____ Period: _____

3-6: Spot the Cubes, Then Factor

TEKS P.5.J: solve polynomial equations by applying a variety of techniques. Factoring is the tool. Tags: (B) = Basic (recognize), (I) = Intermediate (recognize and factor correctly).

Warm-up: Is it a cube?

1. Fill in the perfect cubes. (B)

n	1	2	3	4	5	6	7	8	9	10
n^3	1									

2. Is each a perfect cube? If yes, write it as (_____)^3. If no, write N. (B)

x^3	**x^6**	**$8x^3$**	**$27y^9$**	**$64a^3b^3$**	**$9x^3$**	**x^8**
$(x)^3$						

Explore, Part A: Multiply and notice

Multiply and simplify each product.

Product	**Expanded (simplified)**
$(x - 1)(x^2 + x + 1)$	
$(x + 1)(x^2 - x + 1)$	
$(x - 3)(x^2 + 3x + 9)$	
$(x + 3)(x^2 - 3x + 9)$	

What happened to the middle terms in every row? What do all four answers have in common?

Explore, Part B: The same thing with letters

Multiply **$(A - B)(A^2 + AB + B^2)$**. Write all six products, then combine like terms.

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Now **predict**: $(A + B)(A^2 - AB + B^2)$ = _____. Check by multiplying on the lines.

Use Parts A and B to fill in both patterns.

Pattern	Factored form
$A^3 + B^3$ =	(_____)(_____)
$A^3 - B^3$ =	(_____)(_____)

S-O-A-P: In the binomial, the sign is the _____ as the original. The middle sign of the trinomial is the _____. The last term of the trinomial is always _____.

Practice

Factor completely (take out the GCF first, then use the pattern), then check by multiplying back. Leave the trinomial factor alone when you are done.

3. Sort. Write SC (sum of cubes), DC (difference of cubes), DS (difference of squares), or N (neither). For SC or DC, name A and B. (B)

Expression	Type	A	B
(a) $x^3 - 216$			
(b) $8x^3 + 125$			
(c) $x^3 + 9$			
(d) $x^2 - 64$			
(e) $27y^6 - 1$			
(f) $16x^3 + 1$			

4. $x^3 - 64$ (I)

5. $x^3 + 1000$ (I)

6. $27x^3 + 64$ (I)

7. $125x^3 - 8$ (I)

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8. $m^3 + 64n^3$ (I)

9. $2x^3 - 250$ (I)

10. $16x^3 + 54$ (Look for the GCF first, then name A and B.) (I)

11. Solve $x^3 + 8 = 0$. (I)

(a) Factor the left side, set each factor equal to zero, and find the **real** solution. Why does the trinomial factor give no real solution? (Use $b^2 - 4ac$.)

(b) Optional preview: after the quadratic formula, you will find the other two (non-real) solutions. Leave blank for now.

12. Find the error. Sam factored $8x^3 + 1$ as $(2x + 1)(4x^2 + 2x + 1)$. (I)

(a) Multiply Sam's answer. Does it equal $8x^3 + 1$? (b) Write the correct factorization. (c) Sam wants to factor $4x^2 - 2x + 1$ again. Use $b^2 - 4ac$ to explain why Sam can stop.

13. A cube-shaped block of foam has edge length **x** centimeters. A small cube-shaped corner with edge 4 cm is cut out and thrown away. (I)

(a) Write the leftover volume V(x) and factor it.

(b) Use your factored form to find V(10). Check it a second way.

(c) Why must x be greater than 4?

Math Time Machine: Around 1918 to 1919, Ramanujan told Hardy that 1729 is the smallest number that is a sum of two cubes in two different ways: $1^3 + 12^3 = 9^3 + 10^3$. "Doubling the cube" with only a compass and straightedge was proved impossible in 1837. A Rubik's Cube has $3 \times 3 \times 3 = 27$ small cubes.